ADVANCED ENGINEERING · PRINCIPAL ARCHITECT · AI

My Student Coach Evidence Packet

Product ownership, technical originality, product history, AI cost claims, adoption & impact, and independent validation — assembled from the Student AI Hub repository.

Product: My Student Coach (Student AI Hub)

Repository: kaparthi86/Student-AI-Hub

Packet date: November 8, 2026 (America/Los_Angeles)

Source folder: docs/principal-architect-ai/

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3. **cost-model.csv** `04-ai-cost-claims/cost-model.csv`

Placeholders marked [[LIKE_THIS]] require production exports (invoices, user counts, letters) before external quantified claims.

README.md

Principal Architect / Advanced Engineering (AI) — Evidence Pack

Product: My Student Coach (Student AI Hub)
Role package: Advanced Engineering · Principal Architect · AI Product Ownership
Repository: `kaparthi86/Student-AI-Hub`
Pack date: 2026-11-08 (America/Los_Angeles)
Repo created: 2026-05-02 · Active through 2026-11-08

This folder is an interview / promotion / portfolio evidence kit. Each category maps to a common hiring bar for senior AI product-and-platform leaders.

Category	Folder	What you will find
Product ownership	`01-product-ownership/`	PRD, roadmap, tickets, launch plan, ADRs
Technical originality	`02-technical-originality/`	Architecture, AI workflows, prompts, model choices, repo map
Product history	`03-product-history/`	Version history, release notes, demo script, screenshot placeholders
AI cost claims	`04-ai-cost-claims/`	Cost methodology, cache savings model, CSV templates, before/after
Adoption and impact	`05-adoption-and-impact/`	Metrics framework, SQL/analytics, feedback loop, testimonials
Independent validation	`06-independent-validation/`	Validation plan, letter templates, talk outline

How to use this pack

1. **Technical originality + product ownership** are ready to present as written; they are grounded in the live codebase and merged PRs.
2. **Cost / adoption / validation** include executable frameworks plus **clearly marked placeholders** for production exports (HF invoices, Supabase auth counts, letters). Replace `[[PLACEHOLDER]]` fields before external sharing.
3. Keep secrets out of this folder. Never paste API tokens, invoices with account numbers, or PII into git.

One-line thesis

Built and shipped a student-first AI coach end-to-end: product wedge, response contracts, Hugging Face Inference routing with response/prompt caching, honor-code study modes, PWA/Android-simple shell, and a conversational Voice Tutor — with measurable cost and quality loops.

Suggested interview narrative (10 minutes)

1. Problem → students need focused study AI, not another general chatbot (PRD + roadmap).

2. Architecture → Express + Supabase Auth + HF Router + SSE + cache (architecture diagram).
3. Originality → response contracts, honor code modes, notebook grounding, voice loop (workflows + prompts).
4. Ownership → ADR trail, launch plan, weekly feedback tickets.
5. Economics → cache + voiceMode token caps + before/after model (cost claims).
6. Proof → release history, demos, then live metrics/letters once filled.

`MANIFEST.md`

Evidence Manifest — checklist against role rubric

Rubric item	Covered by	Ready to present?
PRDs	`01-product-ownership/PRD-my-student-coach.md`	Yes
Product roadmap	`01-product-ownership/product-roadmap.md`	Yes
Tickets	`01-product-ownership/tickets-backlog.md`	Yes
Launch plans	`01-product-ownership/launch-plan.md`	Yes
Decision records	`01-product-ownership/decision-records/*`	Yes
Architecture diagrams	`02-technical-originality/architecture-overview.md`	Yes
AI workflow designs	`02-technical-originality/ai-workflow-designs.md`	Yes
Prompts	`02-technical-originality/prompt-library-export.md`	Yes
Code / repo exports	`02-technical-originality/repository-map.md`	Yes
Model/provider choices	`02-technical-originality/model-provider-choices.md`	Yes
Dated screenshots	`03-product-history/screenshots/README.md`	Needs captures
Demos	`03-product-history/demo-script.md`	Needs recording link
Release notes	`03-product-history/release-notes.md`	Yes
Version history	`03-product-history/version-history.md`	Yes
Provider invoices	`04-ai-cost-claims/invoice-checklist.md`	Needs private PDFs
Usage exports	`04-ai-cost-claims/usage-export-template.csv`	Needs real rows
Cache logs / calcs	`04-ai-cost-claims/cache-savings-model.md`	Needs measurement
Before/after	`04-ai-cost-claims/before-after-comparison.md`	Needs experiment
User counts / analytics	`05-adoption-and-impact/*`	Needs Supabase export
Retention / feedback	`05-adoption-and-impact/feedback-loop.md`	Partial (process yes)
Testimonials	`05-adoption-and-impact/testimonials-template.md`	Needs quotes
Articles / awards / letters	`06-independent-validation/*`	Needs outreach

Score guidance: Technical originality + product ownership are interview-ready from repo truth. Cost, adoption counts, and third-party letters require attaching production exports before claiming quantified impact.

`01-product-ownership/PRD-my-student-coach.md`

PRD — My Student Coach (Student AI)

Field	Value
Document ID	PRD-MSC-2026-01
Owner	Product / Principal Architect (AI)
Status	Living — aligned to shipped main as of 2026-11-08
Primary users	Middle school → college / exam prep students
Success definition	Students complete a study turn (ask → understand → practice) without leaving the product

1. Problem

General-purpose AI chat is abundant and often harmful for schoolwork: long unfocused answers, brand confusion, submission-ready solutions, and weak grounding in the student's own notes. Students need a **study coach** that answers first, teaches the smallest useful next step, respects honor code, and works on phone and laptop.

2. Goals

Priority	Goal	Measurable signal
P0	Fast authenticated access to Ask / Code / Notebook	Time-to-first-answer after Google login
P0	Clear, non-markdown-noisy answers with brand identity	Thumbs-up rate; "**" strip failures = 0
P0	Mobile-simple shell (Home / Tutor / Learning / Profile)	Mobile session completion
P1	Learning labs (Voice, Flashcards, Math, Focus, Gaps)	Lab weekly active use
P1	Conversational Voice Tutor	Turns per voice session ≥ 3
P2	Play-ready Android (TWA/PWA path)	Install / add-to-home rate

3. Non-goals

- Replacing a school LMS or gradebook
- Native Kotlin app before PWA/TWA product-market fit
- General consumer search engine
- Finance / Budget AI (removed — Student AI wedge only)
- Claiming to be ChatGPT / Claude / Perplexity / Google

4. Personas

1. **Exam-focused high schooler** — needs short explanations, quiz mode, flashcards.
2. **College CS student** — needs Code tutor + Practice loop without full homework dumps.
3. **Note-heavy learner** — uploads PDFs/notes; needs Notebook grounded answers.

4. **Voice learner** — wants talk-back study while commuting (Chrome/Edge).

5. Product surface (shipped)

Surface	Job to be done
Home	Orient, resume last study path
Ask (Tutor)	Explain / Guide / Quiz with live-web optional grounding
Code	Patient coding tutor + practice
Notebook	Multi-file analyze + grounded follow-ups
Voice Tutor	Stream → speak-early → auto-listen → barge-in
Flashcards / Math / Focus / Gaps	P1 labs under Learning
Profile	Learner personalization (grade, subjects, explain style)
Login gateway	One-screen Google join + honor / nonprofit disclaimer

6. Constraints

- Open-source / cost-aware inference via Hugging Face Inference Providers
- Supabase Google OAuth; no custom password store
- Honor code: Guide/Quiz must not hand over submittable work
- Privacy: feedback may go to Supabase or local `feedback.ndjson`
- Browser voice APIs only for v1 conversational voice (no realtime native yet)

7. Risks and mitigations

Risk	Mitigation
Model hallucination on homework	Notebook “Not in document”; live-web source chips; honor modes
Inference cost	Response cache TTL, prompt-cache params, voiceMode max tokens
Auth fragility (Safari)	Redirect URL checklist; declared abort controllers; hard-refresh cache bust
Scope creep (hub of hubs)	Wedge: Student AI only; kill Finance / Neo Clouds / Privacy from primary path

8. Launch criteria (public)

- [x] Google login on production URL
- [x] Ask / Code / Notebook streaming paths
- [x] Privacy + Terms pages
- [x] Feedback path (`/api/feedback` + weekly review SQL)
- [x] Mobile bottom nav + Voice conversational loop
- [ ] Production HF usage dashboard linked in cost pack
- [ ] Android Play listing (TWA) after install metrics

9. Open questions

1. Default study mode for new users: Explain vs Guide?
2. Should live web be on by default for Ask?
3. When to move Voice Tutor to a realtime STT/TTS provider?

`01-product-ownership/product-roadmap.md`

Product Roadmap — My Student Coach

Horizon dates are product phases, not calendar estimates. Sequencing reflects merged work May–Nov 2026 and the next engineering slices.

Now (shipped on `main`)

Theme	Outcome	Evidence
Student AI wedge	Hub focused on Ask / Code / Notebook	PR #64 remove Finance AI
Coach Home	Phase-1 home dashboard	PR #68
P1 labs	Voice, Vision, Flashcards, Math, Focus, Gaps	PR #69
UX simplification	Login gateway, teal coach brand, menus	PR #70–#73
Android-simple P0	Bottom nav, personalization, chat controls	PR #74
Conversational voice	Stream + speak-early + barge-in + voiceMode	PR #74
Hardening	Safari `activeChatAbort` declaration	PR #75

Next

Theme	Outcome	Dependencies
Quality loop ops	Weekly thumbs review → one shipped contract fix	Supabase service role + review SQL
Cost dashboard	Export HF usage + cache hit rate into `04-ai-cost-claims`	HF billing API / CSV
Notebook multi-file polish	Stronger source attribution in UI	Existing corpus builder
Practice under topic	Topic-scoped practice from Ask	Practice loop already in Code/Ask
TWA / Play packaging	Trusted Web Activity shell	Stable HTTPS domain + DigiCert

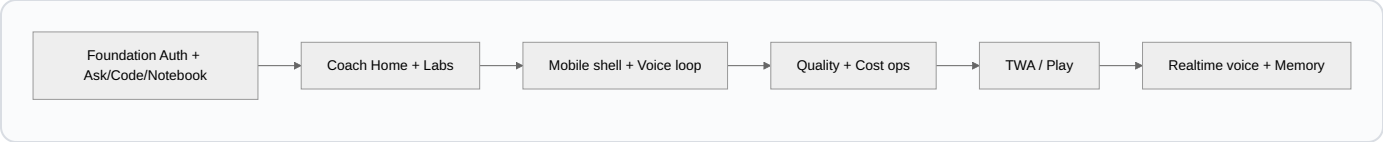
Later

Theme	Outcome
Native realtime voice	Provider STT/TTS or WebRTC tutor
Teacher / classroom mode	Shared decks, anonymized class gaps
Memory vault (opt-in)	Cross-session learner memory with explicit consent
Workspace connectors	Google Drive notes → Notebook

Explicitly not on roadmap

Finance AI, Neo Clouds marketplace inside Student Hub, Privacy Agent as primary Hub tile.

Roadmap diagram



`01-product-ownership/tickets-backlog.md`

Tickets / Backlog — Principal Architect evidence

Ticket IDs are portfolio-stable references. Link each to a GitHub PR or issue when filing in production trackers.

Epic A — Product ownership & launch

ID	Title	Type	Status	Linked work
MSC-101	PRD for My Student Coach public wedge	Spec	Done	`PRD-my-student-coach.md`
MSC-102	Kill Finance AI from Hub	Scope	Done	PR #64
MSC-103	One-screen Google login gateway	UX	Done	PR #73
MSC-104	Privacy / Terms / nonprofit disclaimer parity	Legal UX	Done	legal pages + login copy
MSC-105	Public launch checklist on Render + custom domain	Launch	Done	README deploy section
MSC-106	Weekly feedback review ritual	Ops	Open	`supabase/feedback_weekly_review.sql`

Epic B — Learning quality

ID	Title	Type	Status	Linked work
MSC-201	Response contracts for Ask / Code / Notebook	Prompt eng	Done	`server.js` contracts
MSC-202	Strip leftover `**` from model replies	Quality	Done	PR #72
MSC-203	Explain / Guide / Quiz study modes + honor code	Feature	Done	chat studyMode
MSC-204	Assistant controls: simpler / steps / example / quiz / regen	Feature	Done	PR #74
MSC-205	Learner profile injected into system prompt	Personalization	Done	PR #74
MSC-206	Live web grounding with source chips	Feature	Done	Tavily/Brave/Serper

Epic C — Platform & mobile

ID	Title	Type	Status	Linked work
MSC-301	Coach Home Phase 1	Feature	Done	PR #68
MSC-302	P1 labs (Voice, Vision, Cards, Math, Focus, Gaps)	Feature	Done	PR #69
MSC-303	Android-simple bottom nav	Feature	Done	PR #74
MSC-304	Conversational Voice Tutor loop	Feature	Done	PR #74
MSC-305	Fix Safari undeclared `activeChatAbort`	Bug	Done	PR #75
MSC-306	TWA Play packaging	Infra	Open	—

Epic D — Cost & reliability

ID	Title	Type	Status	Linked work
MSC-401	Response completion cache (TTL + max entries)	Cost	Done	`HF_RESPONSE_CACHE_*`
MSC-402	Prompt-cache parameter forwarding	Cost	Done	`HF_FORWARD_PROMPT_CACHE_PARAMS`
MSC-403	voiceMode short replies + 280 max tokens	Cost/UX	Done	PR #74
MSC-404	Export monthly HF invoice + usage CSV into evidence pack	Ops	Open	`04-ai-cost-claims/`
MSC-405	Cache hit-rate log dashboard	Observability	Open	`HF_LOG_PROMPT_CACHE_USAGE`

Weekly operating ticket template

```
 Title: Feedback week of YYYY-MM-DD
1. Pull GET /api/feedback-summary?days=7
2. Top 2 (reason, mode) pairs
3. One contract or UX change
4. Ship PR; note before/after thumbs in adoption pack
```

`01-product-ownership/launch-plan.md`

Launch Plan — My Student Coach

Field	Value
Document ID	LAUNCH-MSC-2026-PUBLIC
Environments	Local → Render → Custom domain
Auth	Supabase Google OAuth
Inference	Hugging Face Inference Providers (Router)

1. Pre-launch gate

Gate	Owner check	Pass criteria
Secrets	Host env only	`HF_API_TOKEN`, Supabase keys not in git
Auth URLs	Supabase + Google Cloud	Site URL + Redirect URLs + JS origins match HTTPS host
Health	`/api/health`	`hfConfigured: true`, expected `hfModel`
Legal	Privacy + Terms live	Linked from login and footer
Beta banner	Env	`BETA_TESTING=0` for public; `=1` for invite cohort
Feedback	Optional	Service role + `assistant_feedback` table **or** accept ndjson
Cold start	Hosting plan	Accept free-tier ~30s or upgrade before campaign

2. Launch sequence

1. Merge to `main`; confirm Render auto-deploy.
2. Verify custom domain TLS.
3. Smoke: Google login → Ask stream → Code → Notebook upload → Voice one turn.
4. Share **one** canonical HTTPS URL.
5. Day-0: watch auth failures and HF 401/model_not_supported.
6. Day-7: feedback summary + cost export into evidence pack.

3. Comms kit

- One-liner: “My Student Coach — focused AI for studying, not another chatbot.”
- Demo script: see `../03-product-history/demo-script.md`
- Investor leave-behind: `pitch/index.html` (print to PDF)

4. Rollback

Failure	Action
HF auth outage	Demo mode message; status toast; keep login/static UX

Failure	Action
OAuth break	Re-check Redirect URLs; temporary `BETA_TESTING=1`
Bad model routing	Change `HF_MODEL` to known-good `:fastest` / `:groq` suffix without code deploy if env-only

5. Post-launch 30-day plan

Week	Focus
1	Stability: auth, abort/stop, Safari
2	Quality: thumbs reasons → one contract fix
3	Cost: cache hit rate + voiceMode savings snapshot
4	Mobile: TWA spike or install prompt A/B

`01-product-ownership/decision-records/ADR-001-hf-inference-router.md`

ADR-001 — Hugging Face Inference Providers (Router) as primary LLM path

Field	Value
Status	Accepted
Date	2026-05 (established) · reaffirmed 2026-11
Deciders	Principal Architect / Founding engineer

Context

The product needs chat completions (streaming + JSON), optional vision, and cost control without operating GPUs.

Decision

Use Hugging Face Router `https://router.huggingface.co/v1/chat/completions` with env-selected Hub model IDs that include a routing suffix (e.g. `deepseek-ai/DeepSeek-V4-Pro:fastest`). Optional `HF_MODEL_VISION` for image turns.

Consequences

- + OpenAI-compatible payload shape; easy SSE streaming
- + Provider flexibility via Hub routing suffixes (`:fastest` , `:groq` , …)
- + Aligns with student/nonprofit cost posture
- – Model availability depends on enabled Inference Providers
- – Operational debugging for 401 / model_not_supported must be first-class (implemented in `server.js`)

`01-product-ownership/decision-records/ADR-002-student-ai-only-wedge.md`

ADR-002 — Student AI only product wedge

Field	Value
Status	Accepted
Date	2026-11-06
Deciders	Product + engineering
Evidence	PR #64 (Finance AI removed); earlier Hub experiments (Privacy Agent, Neo Clouds) demoted

Context

The Hub accumulated adjacent experiments (Finance AI, Privacy Agent, Neo Clouds). This diluted the student-study promise and increased UX/auth surface.

Decision

Primary product is **My Student Coach / Student AI** (Ask, Code, Notebook + learning labs). Other experiments may live as separate repos or archived folders, not Hub tiles.

Consequences

- + Clear story for users, investors, and Play listing
- + Faster UX consistency (teal coach brand, single home)
- − Abandoned short-term Finance AI usage
- − Requires discipline against new hub-of-hubs ideas

`01-product-ownership/decision-records/ADR-003-response-and-prompt-cache.md`

ADR-003 — Server response cache + optional prompt-cache params

Field	Value
Status	Accepted
Date	2026
Code	`completionCacheGet/Set` , `HF_RESPONSE_CACHE_TTL_SEC` , `HF_FORWARD_PROMPT_CACHE_PARAMS`

Context

Repeated student questions (same mode + similar history) and notebook follow-ups burn tokens. Providers may also support prompt-prefix caching.

Decision

1. Hash `(model, kind, userKey, messages)` and cache completed JSON/SSE payloads in-process with TTL + max entries.
2. Skip cache for live-web turns (freshness).
3. Optionally forward provider prompt-cache knobs when enabled.
4. Prefer static system prefixes (Notebook static system) for prefix-cache friendliness.

Consequences

- + Lower HF bill and latency on repeats
- + Auditable via usage + hit logs for cost claims pack
- − In-process cache does not share across multi-instance hosts (acceptable for MVP; Redis later)
- − Stale answers if TTL too high — tune with `HF_RESPONSE_CACHE_TTL_SEC`

`01-product-ownership/decision-records/ADR-004-pwa-before-native.md`

ADR-004 — PWA / Android-simple shell before native Kotlin

Field	Value
Status	Accepted
Date	2026-11
Evidence	`android-shell.css`, bottom nav, webmanifest, PR #74

Context

Play-ready presence is desirable, but a full Kotlin rewrite would freeze product learning velocity.

Decision

Ship an **Android-simple web shell** (bottom nav Home / Tutor / Learning / Profile, large tap targets, conversational Voice Tutor on Web Speech APIs) and treat **TWA** as the Play packaging path. Defer native realtime voice and Kotlin UI until install/retention justify it.

Consequences

- + One codebase; fast iteration
- − Voice quality limited by browser STT/TTS
- − Some Safari Web Speech gaps — document Chrome/Edge as best voice path

`01-product-ownership/decision-records/ADR-005-voice-conversational-loop.md`

ADR-005 — Almost-conversational Voice Tutor (stream + speak-early)

Field	Value
Status	Accepted
Date	2026-11-08
Evidence	PR #74; `public/coach-labs.js` ; `voiceMode` in `server.js`

Context

Turn-based “listen → wait for full answer → speak all” feels like a walkie-talkie, not a tutor.

Decision

1. Stream `/api/chat` with `voiceMode: true`.
2. Speak sentence boundaries as tokens arrive.
3. Auto-relisten after TTS when enabled.
4. Barge-in: mic tap aborts fetch + cancels speech.
5. Cap spoken length and server `max_tokens` (~280) for short replies.

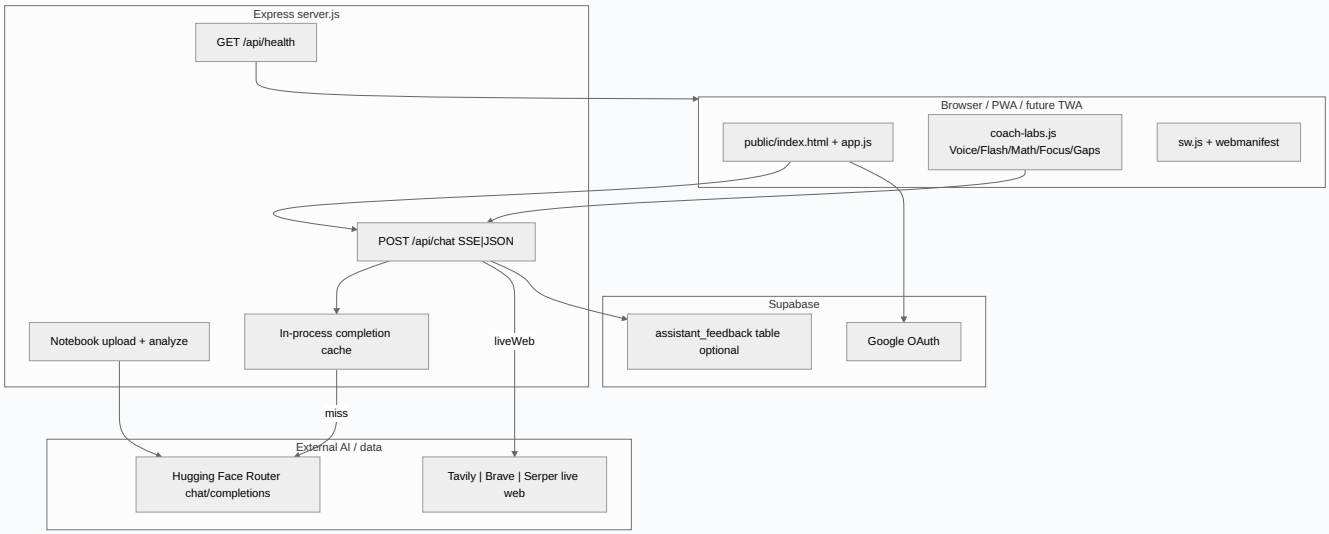
Consequences

- + Perceived latency drops dramatically
- + Lower token spend vs long essay answers
- – Sentence segmentation heuristics can mis-chunk abbreviations
- – Not true full-duplex realtime; native path remains Later roadmap

02-technical-originality/architecture-overview.md

Architecture Overview — My Student Coach

System context



Runtime building blocks

Layer	Choice	Why
UI	Vanilla SPA (no React runtime in production coach)	Fast load, simple deploy, full control of streaming UX
Auth	Supabase Google	Student-friendly SSO; no password DB
API	Node 18+ Express	Single process; easy Render blueprint
LLM	HF Inference Providers	Cost-flexible open models; OpenAI-compatible API
Docs	multer + pdf-parse	Notebook multi-file corpus
Voice	Web Speech STT/TTS + SSE speak-early	Ship conversational feel without native stack
Mobile	CSS Android shell + PWA	Play path via TWA later

Key request path (Ask stream)

1. Client `fetchAuthed('/api/chat', { stream: true, studyMode, learnerProfile, liveWeb? })`
2. Server builds system = identity + response contract + mode style + UI language + learner profile (+ voiceMode / live web appendix).
3. Cache lookup by hash; on miss, HF SSE proxied to client.
4. Client renders tokens; optional Stop via `AbortController` (`activeChatAbort`).
5. Optional thumbs → `/api/feedback` .

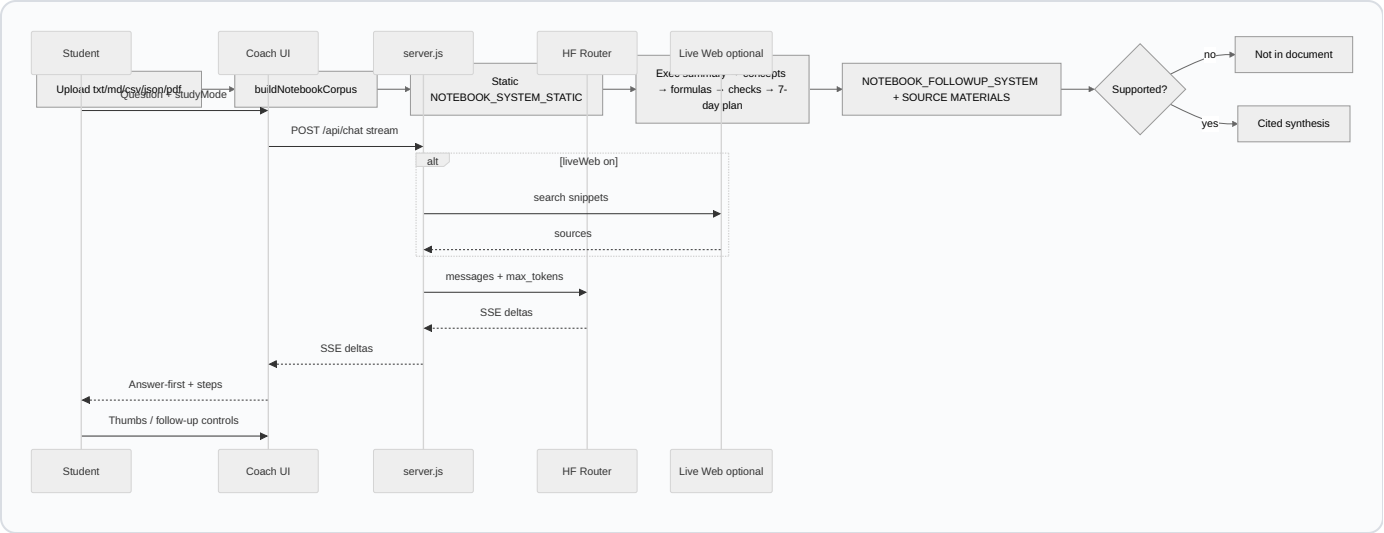
Security / tenancy notes

- HF and Supabase **service** secrets stay on server / host env.
- Frontend only holds Supabase anon key in `public/config.js`.
- OAuth redirect allowlist is a launch gate (Safari pattern errors).
- Honor-code modes constrain assistant behavior in prompt space (not a sandbox).

02-technical-originality/ai-workflow-designs.md

AI Workflow Designs

1. Ask / Learn — teach the smallest useful next step



Design choices: Answer-first contract; Explain/Guide/Quiz styles; optional live web; regenerate and “explain simply” controls without new page.

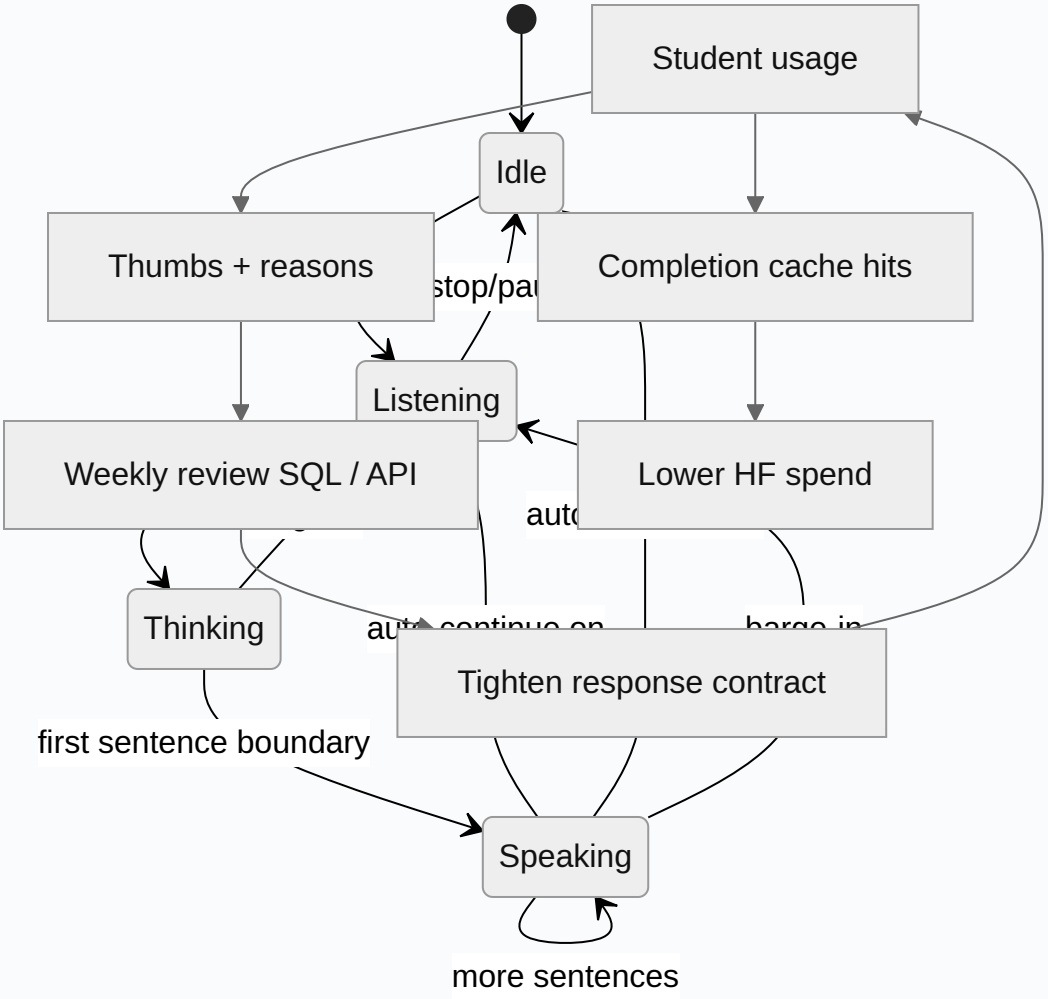
2. Notebook — grounded multi-file study

Design choices: Static system prefix for cache friendliness; document-only truth; structured analyze sections; follow-ups cannot invent facts.

3. Code tutor + practice

- System: patient coding tutor + code fences only for code.
- Practice loop: check / skip under replies (shipped under Code; Ask parity).
- Guide mode blocks full submittable solutions.

4. Voice Tutor — almost conversational



Design choices: `voiceMode` short replies; speak-early; barge-in aborts fetch + TTS session id; Chrome/Edge preferred.

5. Quality / cost control loop

`02-technical-originality/prompt-library-export.md`

Prompt Library Export (from `server.js`)

Exported for originality / IP evidence. Source of truth remains the repository. Do not treat this file as the runtime config.

Brand identity (all chat modes)

> You are My Student Coach. If asked which product or model produced this response, answer: "This answer is from My Student Coach." ... Never claim to be Perplexity, ChatGPT, Claude, Google, or any other brand.

Ask contract (summary)

1. Direct answer first (1–2 sentences).
2. Smallest useful explanation (~180 words default unless depth requested).
3. Numbered steps for sequences; hyphen bullets for lists.
4. Call out one misconception when relevant.
5. No filler openers; no `**` emphasis.

Code contract (summary)

Patient coding tutor; Markdown fences for code only; numbered procedures; honor Guide/Quiz constraints.

Math & Science contract (summary)

Numbered reasoning; formulas/units; common mistakes line; one similar practice prompt.

Study mode styles

Mode	Instruction intent
Explain	Full response contract
Quiz	One check question; don't reveal answer early; honor code
Guide	Hints/questions only; no final answer / submittable code

Voice mode

Keep entire answer under ~80 words; 2–4 short spoken sentences; no markdown/lists; end with one short follow-up question; server caps ~280 `max_tokens`.

Notebook analyze (structured)

Exact `##` sections in order:

1. Executive summary
2. Key concepts
3. Formulas and procedures
4. Practice checks (exactly 5)
5. 7-day study plan

Rule: use only source materials; else “Not in document”.

Notebook follow-up

Document-grounded; start with direct answer; no invented citations; identity line as My Student Coach.

Learner profile injection

When present, include grade level, subjects, and explain style (`balanced` | `simple` | `steps` | `examples`) so the model personalizes without storing long memory server-side by default.

Live web appendix

When live web is requested and configured, append retrieved snippets and require grounding + source-aware answering (client shows source chips).

`02-technical-originality/model-provider-choices.md`

Model & Provider Choices

Primary path

Knob	Default / pattern	Role
`HF_CHAT_URL`	`https://router.huggingface.co/v1/chat/completions`	OpenAI-compatible router
`HF_MODEL`	`deepseek-ai/DeepSeek-V4-Pro:fastest` (env-overridable)	Text + chat
`HF_MODEL_VISION`	optional Hub id with routing suffix	Image turns
Routing suffix	`:fastest`, `:groq`, etc.	Required by Inference Providers

Why Hugging Face Inference Providers

1. **Cost leverage** — open models; switch providers via suffix without rewriting the app.
2. **Operational fit** — single token (`HF_API_TOKEN`) with Inference permissions.
3. **Streaming** — SSE compatible with the coach UI stop/abort UX.
4. **Student mission** — avoids locking the product identity to a closed consumer brand.

Ancillary providers

Capability	Providers	Notes
Live web	Tavily, Brave Search, Serper	First configured key wins path in code
Auth	Supabase + Google	Not an LLM vendor
Hosting	Render (`render.yaml`)	Cold start consideration on free tier

Selection rubric (Principal Architect)

Criterion	Weight	DeepSeek-via-HF	Closed consumer API
$/1M tokens flexibility	High	Strong	Weak–medium
Brand independence	High	Strong	Conflict with “My Student Coach”
Latency control	Medium	Good with cache + stream	Good
Vision option	Medium	Via `HF_MODEL_VISION`	Usually built-in
Compliance storytelling	Medium	Open-weight narrative	Vendor DPA dependent

Change control

Model changes are **env-first** when possible. Document any default ID change in release notes and re-run cost before/after spreadsheet.

`02-technical-originality/repository-map.md`

Repository Map / Export Index

Repo: `kaparthi86/Student-AI-Hub` · Language: JavaScript · Created 2026-05-02

Core runtime

Path	Responsibility
`server.js`	Express API, prompts, cache, HF proxy, notebook, feedback
`public/app.js`	Coach SPA: auth, Ask/Code/Notebook, speech, tabs, personalization
`public/index.html`	Shell markup, panels, login gateway
`public/coach-labs.js`	Voice / Flashcards / Math / Focus / Gaps
`public/coach-labs.css`	Lab styles + voice states
`public/android-shell.css`	Bottom nav / mobile shell
`public/simplify-coach.css`	Coach brand simplification
`public/sw.js` + `manifest.webmanifest`	PWA
`public/config.js`	Supabase anon config (no secrets)
`.env.example`	Server secret template
`render.yaml`	Deploy blueprint
`supabase/*.sql`	Feedback schema + weekly review
`pitch/`	Investor deck

Evidence pack (this tree)

`docs/principal-architect-ai/**`

How to export a code snapshot for reviewers

```
git archive --format=zip --output=student-ai-hub-src-$(date +%Y%m%d).zip HEAD \
  server.js public supabase pitch README.md package.json render.yaml \
  docs/principal-architect-ai
```

Exclude `.env`, `node_modules`, and any invoice PDFs with account numbers.

`03-product-history/version-history.md`

Version History

Dates from git / GitHub (UTC in forge; shown as calendar dates). Product naming evolved from Student AI Hub → My Student Coach.

Version tag (logical)	Date	Milestone
0.1 Hub MVP	2026-05-02	Repository created — auth + student AI prototype
0.4 Feedback loop	2026-08-26	Mission feedback loop (PR #44 era)
0.5 Practice	2026-08-28	Practice under Code (PR #45)
0.6 Adjacent bets	2026-08-30 → 09-05	Finance, Privacy Agent, Neo Clouds experiments
0.7 Hub cleanup	2026-09-05 → 09-24	Remove Neo Clouds / return-home Hub
1.0 Student wedge	2026-11-06	Remove Finance AI (PR #64) — Student AI only
1.1 Coach Home	2026-11-06	Phase 1 Home (PR #68)
1.2 Learning labs	2026-11-06	Voice, Vision, Cards, Math, Focus, Gaps (PR #69)
1.3 UX simplify	2026-11-06	Landing, menus, `**` strip, login gateway (PR #70–#73)
1.4 Android-simple + Voice converse	2026-11-08	Bottom nav, personalization, conversational voice (PR #74)
1.4.1 Safari abort fix	2026-11-08	`activeChatAbort` declaration (PR #75)

Semantic versioning note

`package.json` still reports `1.0.0`. Logical product versions above are for portfolio / release communication until npm version is bumped intentionally.

`03-product-history/release-notes.md`

Release Notes

1.4.1 — 2026-11-08 — Safari Ask/Stop fix

- Declared `activeChatAbort` and `activeChatStopBtn` to fix Safari `Can't find variable: activeChatAbort`.
- Cache-bust `app.js`.

1.4.0 — 2026-11-08 — Android-simple + conversational Voice Tutor

- Bottom navigation: Home / Tutor / Learning / Profile.
- Learner profile prefs fed into chat prompts.
- Assistant response controls (explain simply, steps, example, quiz, regenerate).
- Stop generation on Ask streams.
- Voice Tutor: SSE stream, speak-early sentence TTS, auto-relisten, barge-in, `voiceMode` short answers.

1.3.x — 2026-11-06 — Simplify Coach UX

- Single Home surface, teal brand, Coach naming.
- Learn / Profile menus restored.
- Strip leftover `**` from model replies.
- One-screen Google login gateway (+ orphan markup fix).

1.2.0 — 2026-11-06 — P1 learning labs

- Voice, Vision, Flashcards, Math & Science, Focus Mode, Knowledge Gaps.

1.1.0 — 2026-11-06 — Coach Home

- Phase 1 dashboard as primary post-login orientation.

1.0.0 — 2026-11-06 — Student AI only

- Removed Finance/Budget AI from the Hub product path.
- Positioning: focused learning AI.

Earlier 2026

- Auth hardening (Google login / `wireSearchFlow`).
- Hub return-home behavior.
- Experimental Privacy Agent and Neo Clouds (later removed from primary Hub).
- Practice loops and feedback mission work (Aug 2026).

Demo Script — My Student Coach (8–10 minutes)

Audience: hiring panel / investors / design partners
Environment: Chrome or Edge recommended for Voice; production HTTPS URL
Prep: Signed-in test account; one PDF for Notebook; mic permission

Act 1 — Problem (45s)

“Students don’t need another generic chatbot. They need a coach that answers first, respects honor code, and works on a phone.”

Act 2 — Login & Home (60s)

1. Show one-screen Google gateway + disclaimer.
2. Land on Coach Home.
3. Point at Android-style bottom nav.

Act 3 — Ask quality (2 min)

1. Ask a multi-step concept question in **Explain**.
2. Show answer-first, clean steps (no `**` noise).
3. Switch to **Guide** — show hint-only behavior.
4. Use **Explain simply** control on last answer.
5. Optional: Live web on → show source chips.

Act 4 — Notebook grounding (2 min)

1. Upload class notes / PDF.
2. Show structured analyze sections.
3. Ask a follow-up not in the doc → “Not in document”.

Act 5 — Voice loop (2 min)

1. Open Voice Tutor.
2. Ask a short question; note speak-early before full text finishes.
3. Barge in mid-answer; ask a follow-up.
4. Show auto-continue toggle.

Act 6 — Architecture & cost (90s)

1. HF Router + cache + voiceMode token cap.
2. Weekly thumbs → contract fix loop.
3. Close with roadmap: TWA Play, then realtime voice.

Recording checklist

- [ ] Date stamp in filename: `demo-YYYY-MM-DD-msc.mp4`
- [ ] Store under `03-product-history/screenshots/` or private drive (large binaries not required in git)
- [ ] Link path in this file after capture: `[[DEMO_VIDEO_URL]]`

`03-product-history/screenshots/README.md`

Dated screenshots & demos

Git is a poor place for large binaries. Drop files here **or** link to a private drive and keep this index updated.

Date (America/Los_Angeles)	Asset	What it proves	Path / URL
2026-11-08	Login gateway	One-screen Google join	`[[PLACEHOLDER]]`
2026-11-08	Home + bottom nav	Android-simple shell	`[[PLACEHOLDER]]`
2026-11-08	Ask streaming answer	Response contract	`[[PLACEHOLDER]]`
2026-11-08	Notebook analyze	Grounded sections	`[[PLACEHOLDER]]`
2026-11-08	Voice Tutor speaking state	Conversational loop	`[[PLACEHOLDER]]`
2026-11-06	Pre-simplify Hub (optional)	Before/after UX	`[[PLACEHOLDER]]`

Capture tips

- Include OS clock or a dated sticky note in-frame for authenticity.
- Prefer 1× device frames (iPhone + desktop).
- Redact emails / faces of real students.

`04-ai-cost-claims/cost-methodology.md`

AI Cost Claims — Methodology

Principle

Cost claims for interviews or boards must be **reproducible**: invoice → usage export → calculation sheet → architecture lever (cache, max_tokens, model).

Sources of truth

Artifact	Where it comes from	Stored in git?
Provider invoices	Hugging Face billing PDF/CSV	No (sensitive) — store path in checklist
Usage export	HF usage dashboard / API	Template CSV only
Cache logs	Server logs when `HF_LOG_PROMPT_CACHE_USAGE` enabled; completion cache hit counters	Aggregate stats OK
Calculation spreadsheet	`cost-model.csv` + your filled copy	Template yes; filled copy optional/private
Before/after	Same workload scripted against two configs	Yes (methodology + sample)

Unit economics model

```
 Monthly_AI_cost ≈ Σ ( prompt_tokens_i + completion_tokens_i ) × price_tier_i
Effective_cost_per_study_turn = Monthly_AI_cost / study_turns
Cache_savings = tokens_avoided_by_hits × price
Voice_savings ≈ (baseline_completion_tokens - voiceMode_completion_tokens) × voice_turns × price
```

Architectural levers already shipped

1. **Completion response cache** — `HF_RESPONSE_CACHE_TTL_SEC`, `HF_RESPONSE_CACHE_MAX_ENTRIES`
2. **Prompt-cache param forwarding** — `HF_FORWARD_PROMPT_CACHE_PARAMS`
3. **Static Notebook system prefix** — prefix-cache friendly
4. **voiceMode max_tokens ≈ 280** + short-reply instruction
5. **Mode-specific max_tokens** — vision/notebook/live-web caps in `server.js`
6. **Skip cache on live web** — correctness over thrift when freshness matters

Claim language (safe)

- Prefer: “We designed X to reduce completion tokens / avoid repeat HF calls; measured on [date] with [export].”
- Avoid: absolute savings percentages without the CSV attached.

`04-ai-cost-claims/cache-savings-model.md`

Cache & Voice Savings Model

Completion cache (illustrative math)

Replace bracketed numbers with your export.

Metric	Symbol	Example (SAMPLE — replace)
Ask turns / month	T	[[T=12000]]
Repeat hash hit rate	H	[[H=0.18]]
Avg tokens / turn (prompt+completion)	K	[[K=900]]
Price / 1M tokens	P	[[P=USD_x]]
Avoided tokens	T×H×K	—
Monthly cache savings	avoided × P / 1e6	—

Voice mode (illustrative)

Metric	Baseline Ask	Voice mode	Notes
Target answer length	~180 words	~80 words	Prompt contract
max_tokens cap	up to ~720	~280	`server.js`
Expected completion token ratio	1.0	~0.35–0.55	Measure; don't invent

Before / after experiment protocol

1. Freeze model ID.
2. Run 30 scripted prompts in Ask (no live web) with cache cold, then warm.
3. Run 30 Voice Tutor prompts with `voiceMode`.
4. Export token usage; fill `usage-export-template.csv`.
5. Record results in `before-after-comparison.md`.

SAMPLE narrative (label clearly when presenting)

> SAMPLE ONLY: If hit rate is 18% on 12k turns at 900 tokens and $0.50 / 1M tokens, avoided cost ≈ 12k × 0.18 × 900 / 1e6 × 0.50 ≈ $0.97 — small absolute dollars at prototype scale, but the **mechanism** is what scales with users. Replace with real invoice math.

`04-ai-cost-claims/before-after-comparison.md`

Before-and-After Cost Comparison

Field	Value
Experiment window	[[START]] → [[END]]
Model held constant	[[HF_MODEL]]
Owner	[[NAME]]

Config A — Before

- Response cache TTL: [[0 or low]]
- voiceMode: off for voice path
- avg max_tokens Ask: [[ ]]

Config B — After

- Response cache TTL: [[sec]]
- voiceMode: on for Voice Tutor
- voice max_tokens: 280

Results

Metric	A	B	Δ
Total prompt tokens	[[ ]]	[[ ]]	[[ ]]
Total completion tokens	[[ ]]	[[ ]]	[[ ]]
Cache hit rate	[[ ]]	[[ ]]	[[ ]]
Est. $ (from invoice rate)	[[ ]]	[[ ]]	[[ ]]
p50 time-to-first-token	[[ ]]	[[ ]]	[[ ]]
Voice mean spoken chars	[[ ]]	[[ ]]	[[ ]]

Attached evidence

- Invoice file (private): `[[PATH]]`
- Usage CSV: `[[PATH]]`
- Notes: `[[ ]]`

`04-ai-cost-claims/invoice-checklist.md`

Provider Invoice & Usage Checklist

Do **not** commit real invoices with account numbers to a public repo.

- [ ] Download Hugging Face invoice PDF for [[MONTH]]
- [ ] Export usage CSV → fill `usage-export-template.csv`
- [ ] Confirm `HF_MODEL` / router URL matches production `/api/health`
- [ ] Capture cache env: `HF_RESPONSE_CACHE_TTL_SEC`, max entries, prompt-cache flags
- [ ] Screenshot billing portal header with date (redact account id)
- [ ] Store private pack path: `[[SECURE_DRIVE_PATH]]`
- [ ] Cross-link totals into `before-after-comparison.md`
- [ ] Optional: Render host invoice (compute ≠ model tokens — keep separate)

`05-adoption-and-impact/metrics-framework.md`

Adoption & Impact — Metrics Framework

North-star

Study turns completed per week — a user sends a learning request and receives a usable coach reply (Ask / Code / Notebook / Voice).

Metric tree

Level	Metric	Source
Acquisition	Sign-ups / week	Supabase Auth users
Activation	First successful Ask within 24h of signup	App analytics or server logs
Engagement	WAU, study turns / WAU	Auth + `/api/chat` logs
Quality	Thumbs-up rate; top negative reasons	`assistant_feedback`
Retention	D1 / D7 / D30 return	Auth last_sign_in / events
Mobile	Bottom-nav sessions; Voice sessions ≥3 turns	Client events (to instrument)
Cost	$ / study turn	HF export ÷ turns

Current instrumentation in product

Already available	Gap to close
Supabase Auth user list	Funnel dashboard
`/api/feedback` + weekly SQL	Automated weekly email
`/api/health` model config	Public status page
Client local histories	Server-side anonymized event table

Portfolio numbers

Until production exports are attached, present **methodology + qualitative impact** (PRs shipped, demo, feedback loop design). Fill:

Metric	Value	As-of date	Export
Registered users	[[N]]	[[DATE]]	Supabase
WAU	[[N]]	[[DATE]]	[[ ]]
Study turns / week	[[N]]	[[DATE]]	[[ ]]
Thumbs-up %	[[N]]	[[DATE]]	feedback summary
D7 retention	[[N]]	[[DATE]]	[[ ]]

`05-adoption-and-impact/analytics-queries.md`

Analytics Queries

Supabase — feedback (shipped SQL companions)

Use repo files:

- `supabase/assistant_feedback.sql` — schema
- `supabase/feedback_weekly_review.sql` — weekly negatives

API equivalents while signed in:

- `GET /api/feedback-summary?days=7`
- `GET /api/feedback-review?days=7` (founder emails via `FEEDBACK_REVIEW_EMAILS`)

Supabase — user counts (run in SQL editor)

```
 -- Registered users (Auth schema; permissions vary by project)
select count(*) as users
from auth.users;

select date_trunc('week', created_at) as week, count(*) as signups
from auth.users
group by 1
order by 1 desc
limit 12;
```

Suggested event table (not yet required in app)

```
 create table if not exists public.product_events (
  id bigserial primary key,
  user_id uuid,
  event_name text not null,
  mode text,
  created_at timestamptz default now(),
  meta jsonb default '{}'::jsonb
);

-- Example: study_turn_completed, voice_barge_in, notebook_analyze
```

Instrument later from `sendChatMessage` success path and Voice Tutor completion.

`05-adoption-and-impact/feedback-loop.md`

Customer Feedback Loop

System

1. User taps Helpful / Not helpful on assistant replies.
2. Server writes Supabase `assistant_feedback` **or** `feedback.ndjson`.
3. Weekly: review top `(reason, mode)` pairs.
4. Ship **one** contract or UX fix.
5. Record impact in adoption pack.

Operating cadence

Day	Action
Mon	Pull 7-day summary
Tue	Reproduce top failure in Ask/Code/Notebook
Wed–Thu	Prompt/UX PR
Fri	Note thumbs delta; update release notes

Example closed loops (evidence)

Period	Signal	Change	PR
2026-11	Markdown `**` noise	Strip / polish replies	#72
2026-11	Learn/Profile dead	Menu panel wiring	#71
2026-11	Safari abort crash	Declare abort vars	#75
2026-11	Voice felt turn-based	Conversational loop	#74

Testimonial capture

See `testimonials-template.md`. Prefer written permission + role (student / parent / teacher) without grades PII.

`05-adoption-and-impact/testimonials-template.md`

Testimonials & Customer Feedback Templates

Permissioned quote template

```
 Name / handle: [[ ]]
Role: Student | Parent | Teacher | Advisor
School / org (optional): [[ ]]
Date: [[YYYY-MM-DD]]
Permission to share publicly: Yes / No
Quote:
“[[ ]]”
Context (which feature): Ask | Code | Notebook | Voice | Other
```

In-product feedback export fields

`rating`, `reason`, `mode`, `study_mode`, `created_at` — from `assistant_feedback`.

Synthesis slide (for interviews)

- N quotes curated
- Top 3 pain themes before fixes
- Top 3 delights after Coach simplify + Voice

Fill with real voices before external use: `[[TESTIMONIAL_PACK_LINK]]`

`06-independent-validation/validation-plan.md`

Independent Validation Plan

Hiring panels and promotion committees overweight **third-party** proof. This plan lists channels and readiness.

Channel	Status	Artifact to attach
Design-partner schools / clubs	[[STATUS]]	LOI or email thread PDF
Nonprofit / education advisor letter	[[STATUS]]	Signed PDF
Conference talk / meetup	Outline ready	`conference-talk-outline.md`
Written articles / blogs	[[STATUS]]	URL + PDF mirror
Awards / accelerators	[[STATUS]]	Certificate
Expert technical review	[[STATUS]]	Architecture review note
Press	[[STATUS]]	Link

Near-term validation actions

1. Run 5 moderated student sessions; capture quotes with permission.
2. Ask one educator for a one-page letter (template below).
3. Submit a local meetup talk using the outline.
4. Publish a technical blog: “Response contracts + honor-code modes for student AI.”
5. Keep pitch deck (`pitch/`) current for investor/advisor meetings.

06-independent-validation/letter-templates.md

Letter Templates (Customer / Expert)

A. Educator / advisor letter

```
[Date]

To whom it may concern,

I am [Name], [Title] at [Institution]. I reviewed My Student Coach / Student AI Hub
on [date]. The product's focus on answer-first tutoring, honor-code Guide/Quiz modes,
and document-grounded Notebook study is relevant to [students / classroom use-case].

In our trial, [1–2 factual observations]. I support continued development of this
approach for [reason].

Sincerely,
[Name]
[Contact]
```

B. Technical expert letter

```
[Date]

I reviewed the architecture of My Student Coach (Express + Supabase Auth + Hugging Face
Inference Router, SSE streaming, completion caching, and conversational Voice Tutor).
Notable engineering choices include [cache strategy / response contracts / voiceMode].

Based on the repository and demo on [date], the system demonstrates principal-level
ownership of AI product architecture and cost-aware inference design.

[Name], [Role]
```

C. Student permissioned testimonial (short)

```
I used My Student Coach for [task]. It helped me [outcome] without [pain of generic chatbot].
— [First name], [Grade/Year], [Date]
```

Store signed PDFs privately; link here: `[[LETTERS_FOLDER]]`

Conference / Meetup Talk Outline

Title: Building a Student AI Coach: Response Contracts, Honor Codes, and Cost-Aware Inference
Length: 25 minutes + 5 Q&A
Audience: AI engineers, edtech builders, principal architects

Outline

1. **Hook (2m)** — Generic chat fails schoolwork; brand confusion; submission-ready answers.
2. **Product wedge (3m)** — My Student Coach: Ask / Code / Notebook only.
3. **Response contracts (5m)** — Answer-first; strip markdown noise; Guide/Quiz honor modes.
4. **Grounding (4m)** — Notebook “Not in document”; optional live web sources.
5. **Architecture (5m)** — HF Router, SSE, abort/stop, completion cache, voice speak-early.
6. **Economics (3m)** — Cache + voiceMode token caps; measure $/study turn.
7. **Close (3m)** — PWA→TWA path; what we’d do with realtime voice next.

Materials to bring

- Architecture mermaid from this pack
- 90-second Voice demo video
- One before/after answer screenshot
- Cost methodology one-pager (no secrets)

Abstract (150 words)

```
 [[DRAFT]] Students increasingly use general-purpose AI for homework, but generic chat
optimizes for fluency—not learning. My Student Coach is a focused study product that
combines answer-first response contracts, honor-code tutoring modes, document-grounded
notebook workflows, and a cost-aware Hugging Face inference path with caching and a
conversational voice loop. This talk shares the architecture and product decisions
behind shipping that system as a principal-level AI engineer/owner.
```

04-ai-cost-claims/usage-export-template.csv

usage-export-template.csv

```
date,provider,model,request_kind,prompt_tokens,completion_tokens,cached,voice_mode,live_web,study_mode,notes
2026-11-01,huggingface,deepseek-ai/DeepSeek-V4-Pro:fastest,chat_sse,650,420,false,false,false,explain,SAMPLE row — replace
2026-11-01,huggingface,deepseek-ai/DeepSeek-V4-Pro:fastest,chat_sse,650,0,true,false,false,explain,SAMPLE cache hit — completion billed 0 if fully cached
2026-11-08,huggingface,deepseek-ai/DeepSeek-V4-Pro:fastest,chat_sse,480,160,false,true,false,explain,SAMPLE voiceMode
```

04-ai-cost-claims/cost-model.csv

cost-model.csv

```
metric,unit,before_value,after_value,notes
ask_turns_per_month,count,[[ ]],[[ ]],From logs
cache_hit_rate,ratio,[[ ]],[[ ]],Completion cache
avg_prompt_tokens,tokens,[[ ]],[[ ]],Usage export
avg_completion_tokens,tokens,[[ ]],[[ ]],Usage export
voice_avg_completion_tokens,tokens,[[ ]],[[ ]],voiceMode on
price_per_million_tokens,usd,[[ ]],[[ ]],From HF invoice
est_monthly_ai_usd,usd,[[ ]],[[ ]],= turns * tokens * price
est_usd_per_study_turn,usd,[[ ]],[[ ]],
```